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Real-Time Multi-Input Diabetic Retinopathy Stage Detection using EfficientNet-B3 and Fundus Images

Evangelin Maria J
Department of Computer Science and Engineering
Sathyabama Institute of Science and Technology
Chennai, India
Email: evangelinmaria25@gmail.com

Durga Devi G
Department of Computer Science and Engineering
Sathyabama Institute of Science and Technology
Chennai, India
Email: tarahhari07@gmail.com

Prem Jacob T
Department of Computer Science and Engineering
Sathyabama Institute of Science and Technology
Chennai, India
Email:premjac@yahoo.com

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Abstract— Diabetic Retinopathy (DR) is a severe eye disease and one of the main causes of blindness for people with long term diabetes. Early detection and treatment can greatly reduce the risk of vision loss. However, traditional Diabetic Retinopathy diagnosis depends on trained ophthalmologists manually examining retinal fundus images. This method is slow, expensive, and often unavailable in rural and underserved regions. In addition, different doctors may interpret the same image differently, leading to variation in diagnosis and making the test results less accurate and less trustworthy. This proposed system introduces an AI driven Automated Diabetic Retinopathy Detection System that classifies retinal fundus images into five clinical stages: No DR, Mild, Moderate, Severe, and Proliferative Diabetic Retinopathy. The proposed system is built using the EfficientNet-B3 deep learning model, chosen for its high accuracy and processing speed. To ensure fair and consistent performance across all stages, the model is trained on a balanced dataset to reduce class imbalance. To improve transparency and trust in the predictions, Grad-CAM visual maps are generated to highlight the important regions of the retina that influence each decision, helping doctors to understand how the system makes its decisions. The trained model achieves an accuracy of 98.77%, showing that the system can accurately find diabetic retinopathy. Beyond the model, the solution is implemented as a user-friendly web application that allows multiple ways of uploading images such as direct file upload, QR based mobile upload, and real time webcam capture. The platform also include patient profile management, visit history logs, automated PDF report generation, and email based report sharing. The mix of reliable automated technology and an easy interface makes this system suitable for remote eye care and local health centers. Overall, the proposed solution offers a scalable, cost effective, and clinically useful solution that can improve early Diabetic Retinopathy detection, reduce the workload for ophthalmologists, and help to prevent vision loss caused by diabetes.

Keywords— Diabetic Retinopathy, EfficientNet-B3, Fundus Image Analysis, Grad-CAM, Medical Image Classification.

I. INTRODUCTION

Diabetic Retinopathy is a common side effect of diabetes. It's also a reason people go blind when treatment gets delayed [1]. High blood sugar over time harms tiny eye vessels, slowly creating small bulges, bleeding spots, fluid leaks, or new faulty veins these issues harm sight unless caught fast. As more people develop diabetes, Diabetic Retinopathy is becoming more common, which makes early detection very important to protect vision. Studies say about every third diabetic person shows signs of this condition; catching it on time could stop around 9 out of 10 serious vision problems [2]. Still, most don't feel symptoms at first, meaning they visit doctors only once their eyesight worsens which makes recovery tougher. Understanding the structure of the retina helps in identifying how the disease progresses. As shown in Fig. 1, a healthy retina has sharp parts like the macula, fovea, optic nerve head, along with evenly spread blood vessels. On the flip side, Fig. 2 shows Diabetic Retinopathy changing those areas step by step. At first, Diabetic Retinopathy brings small issues like tiny bulges in vessel walls but later it leads to bleeding, yellow deposits, fluffy white patches, plus new but faulty blood vessels. These harmful shifts mess up the retina's structure, plus cause permanent harm unless caught early. Telling sick retinas apart from healthy ones shows why smart tools are needed ones that spot small injuries while telling different Diabetic Retinopathy phases apart.

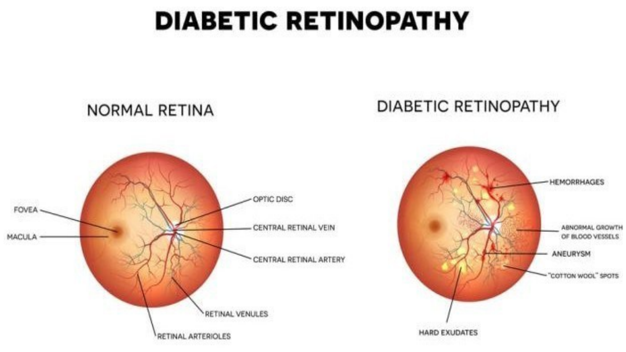

Fig. 1. Structure of a normal retina showing major anatomical regions[14].

Manual checks for diabetic eye issues usually use special tools like retina photos, detailed scans, or wide pupil exams each needing expert doctors, expensive equipment and long clinic hours [3]. They're often slow and expensive; worse, people in remote areas rarely get access because there aren't enough eye experts around [4]. More, human reviews can differ from one doctor to another, especially when telling apart tiny signs like small bulges, fluffy spots, yellow deposits, or slight bleeding. Errors increase when images are of poor quality, such as being grainy, dark, blurry, or

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unevenly lit, which makes large-scale and reliable screening more difficult [5], [6]. With diabetes spreading worldwide, the surge in needed eye tests is straining medical services, pushing a stronger push toward fast, automatic, trustworthy check up methods. In reaction to these limits, tools powered by the computers and smart algorithms are now widely used to analyze eye images. Instead of today's advanced models, older ones like Support Vector Machines or K-Nearest Neighbors depended on manually designed patterns such as LBP or shape based traits; yet they often failed when faced with blurry inputs or new data [7]. With the arrival of deep learning, especially networks called CNNs, spotting diabetic issues in eyes became far more accurate since key details were pulled straight from photos without manual help [1]. Models like VGG16, InceptionV3, and others have been tested on different retina image sets including EyePACS and Messidor and proved effective at telling apart disease stages across sources like DIARETDB and APTOS 2019 [3][8][9].

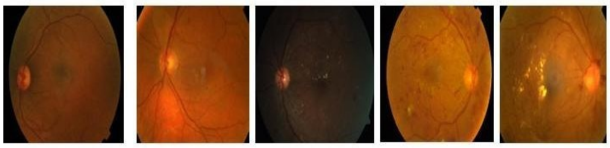

Fig. 2. Five stages Of Diabetic Retinopathy.

Even with progress, big problems remain when building solid Diabetic Retinopathy detection tools. The data skews heavily toward normal or mild cases, leaving rare examples of moderate to advanced stages [4][10]. Because of this lopsided spread, algorithms tend to overlook less common forms, making it harder to catch early signs or tell similar levels apart. Tiny warning signs like microaneurysms are tough for systems to pick up, which messes up accuracy in borderline cases [6] [11]. Additionally, differences in how photo capture is uneven lighting, blurry details, grainy quality, varied cameras weaken reliability and limit use across real world clinics. Most current systems zero in on picture only spotting, skipping patient context that might boost diagnosis quality plus make stage calls more precise [7]. In addition, a bunch of strong CNN setups need heavy computing power so running them live gets tough where gear is weak or phones are used [4][10]. A big issue with today's deep learning is how hard it is to see what's going on inside. CNNs work well, yet they're like closed boxes tough to peek into when decisions are made. In healthcare settings, doctors need clear reasons behind a prediction about eye disease levels [3][9]. Tools from explainable AI, like Grad CAM or LRP, highlight key areas in retina scans so people can spot what influenced the result. That helps build confidence and let human team up better with machines. Research shows giving insight doesn't just boost trust it cuts down wrong calls by checking if damage spots were correctly flagge [6][11]. To tackle these issues, new research suggests updated mixed designs like combined Inception ResNet setups, network types that use graph info with multiple data forms, or deep learning groups built using randomized methods [7][10][11]. These structures aim to boost how features are pulled from lesions of varying sizes, include broader patient specific links, while also lifting accuracy when sorting more than two classes. Still, problems like uneven datasets, working well across varied retina scans, along with live use in clinics haven't been fully sorted out yet. Top of that, plenty of current tools fixate only on building models but skip real world need for instance, ease of use, fitting into doctor routines, or adjusting to distinct screening settings. Because of these limits, this badly need a Diabetic Retinopathy detection tool as both precise and easy to understand, runs fast, doesn't demand heavy computing power, yet works well in hospitals [4]. Models based on EfficientNet are getting popular thanks to their smart scaling method which boosts results while using fewer resources, fitting perfectly for live or phone based use. Top of that, linking these systems with explainable AI tools and simple to use platforms helps turn lab ideas into real world medical solutions.

II. Literature Review

Studies into automatic Diabetic Retinopathy spotting started with basic algorithms, then shifted toward complex neural networks or mixed systems. Every step forward aimed at tackling key medical hurdles like catching tiny damages sooner, dealing with uneven data samples, making models easier to understand, also enabling quick use during check up routines. The first set of research looked at boosting accuracy in spotting lesions using deep neural networks to pull out key features. Because it uses layered convolutional filters, the method in was better at sorting diabetic retinopathy stages by detecting tiny signs like microaneurysms or bleeding spots [1]. The research work tested various deep learning methods finding CNNs worked best since they can pick up clear imag e patterns straight from unprocessed retina scans [2]. At the same time, the team behind argued that high performance isn't enough unless doctors understand how decisions are made, so they added Grad-CAM heatmaps to make their system's outputs more visible and trustworthy [3].

Light ones that can grow easily caught lots of attention lately. A small CNN setup built on EfficientNet-B0, showed good results using way less computing power, so it could work well in low resource spots or handheld tools [4]. At the same time, a look at older and smart tech methods in found machines must help out since doctors aren't always around especially when checking many people fast [5]. A mix method in used vessel maps along with AI to better spot damaged areas, boosting how soon diabetes eye issues get noticed [6]. Researchers looked into more advanced mixed models to boost diabetic retinopathy detection. Instead of just stacking methods, they linked CNNs with graph networks this helped track how eye parts relate in space.

By doing so, the system used both big picture views and structural links to improve severity scoring. Another research sorted out current deep learning approaches, pointing out common roadblocks like inconsistent data quality, poor cross dataset performance, and uneven label distribution. A separate review stressed clarity in decisions not just accuracy and showed how heatmaps can reveal if a model focuses on medically relevant areas. The hybrid setups and mix ups of deep learning models are now solid options instead of one alone networks. A method using randomness built several ways to pull out image traits, which helps handle messy lighting and background fuzz big issues

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in eye scan images [10]. In another case, a combo of Inception and ResNet models showed that teaming up decisions at different steps lifts accuracy in spotting diabetes related vision damage, especially when warning signs look similar across stages [11]. It took a close look at over 26 types of smart algorithms like VGG, DenseNet, MobileNet, ResNet, and EfficientNet and found reusing trained models gives better results detecting eye problems when there aren't many tagged examples [12]. Carrying that idea further, a slimmed down OVO system in used paired classifiers and vote counting to tell apart closely graded disease levels more clearly all while needing less power to run [13]. It can see a similar way of doing things. Like shown in Fig. 1, many tools for spotting Diabetic Retinopathy start off cleaning images. Then they pull out features sometimes old school methods, sometimes deep learning. After that comes sorting results into categories. Some even add visuals to explain their output. This setup find again and again even when models differ in design, how they pick traits, or how they learn.

Though the tested models work well on various Diabetic Retinopathy tests, some problems still exist. A lot of research faces issues with uneven data, especially in mild Diabetic Retinopathy cases tiny abnormalities get missed or wrongfully labeled. When models learn from just one source, they usually don't adapt well to different image types a flaw noted in [8][12]. Even with better explainable AI tools, certain methods give unclear or unreliable heatmaps, so eye doctors can't fully trust automatic results [3][9]. Speed and hardware matter too powerful CNNs need heavy computing power, which blocks use in handheld devices or fast paced clinics but lighter versions might help [4][13]. A key point is most current studies don't mix in real world tools like phone uploads or webcam snapshots, along with handling patient data or auto reports. Instead, they stick to testing just the model part so there's a disconnect between lab results and actual clinic use. Also, even though combined models boost performance, they tend to get harder to manage, making setup, adjustments, and checks tougher in live medical settings.

Still, studies show real gains using deep learning to spot diabetic retinopathy but clear, fast, and practical tools for clinics aren't quite there yet. Issues like catching mild cases, reliable data, phone compatibility, and built in explanations push us to build better screening methods, just like the approach sharing here Fig 3.

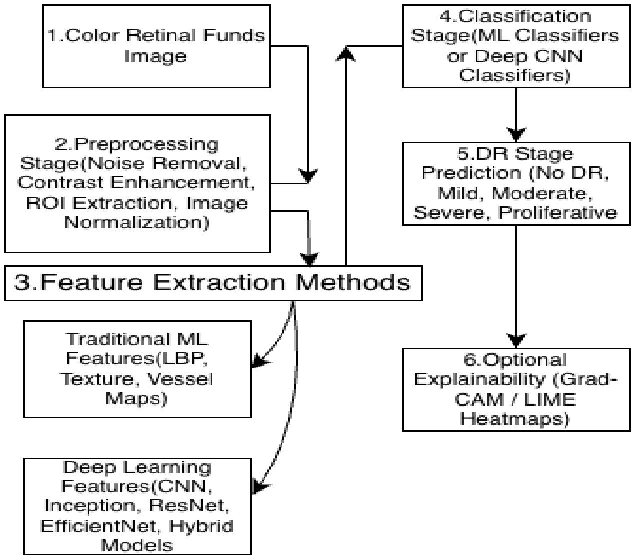

Fig. 3. General process flow of existing diabetic retinopathy detection methodologies.

The diagram shows how automatic diabetic eye diseases detection works first capturing retina images, also clean them up before analysis. After that, key details get pulled out through classic algorithm or modern neural networks instead. These details go into a decision system which labels how the condition is serious.

III. SYSTEM ARCHITECTURE

The suggested Diabetic Retinopathy detection setup works from start to finish using separate parts that fit together deep learning checks eye images while a simple website lets users interact smoothly in actual medical settings. It handles various ways to upload photos, grades condition seriousness without help, shows how decisions are made visually, also keeps reports safe all inside one connected system. The setup splits into separate parts that handle specific jobs keeps things growing smoothly, easier to follow, also simpler to update later Fig 4.

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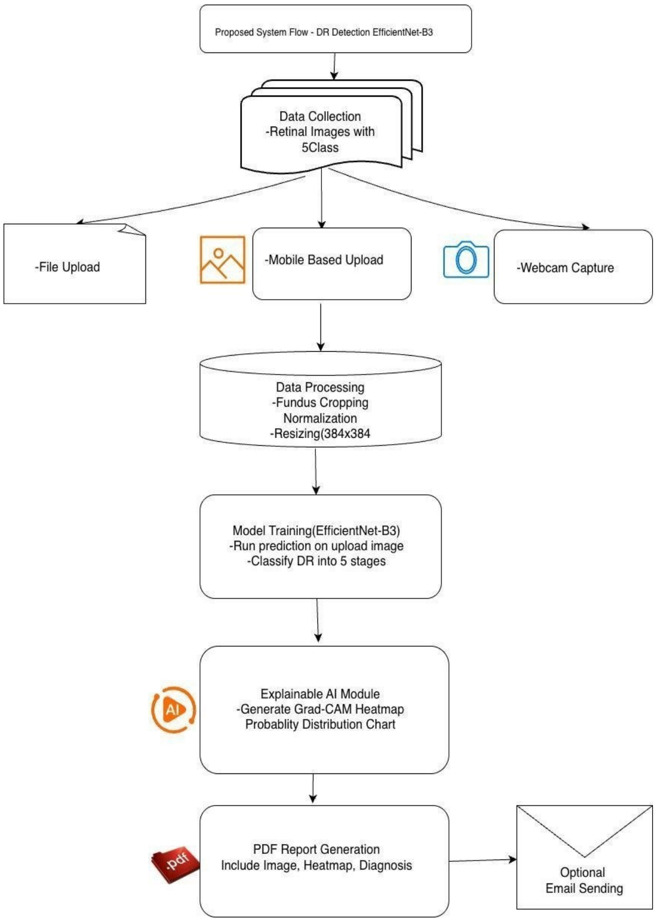

Fig. 4. Architecture of the proposed diabetic retinopathy detection system.

A. *User and Patient Interface Layer*

The User and Patient Interface Layer is where people actually use the system patients, doctors, or tech staff all interact here. Instead of just logging in the system, they get started by building a personal profile online, adding details like age or contact info. When it's time to check past visits, that history pulls up fast thanks to built in tools. To begin an exam, someone might upload a retinal image from their device without extra steps. After analysis, results show clearly on screen alongside visual explanations so users grasp what's happening. Reports come together automatically once findings are confirmed. Whether in clinics, remote consults, or mobile health drives, this setup works smoothly each time in Diabetic Retinopathy detection.

B. *Image Acquisition and Input Layer*

The Image Acquisition Layer handles retinal pictures in different ways users can send files straight from their device, scan a QR code with a phone, or use a built in camera if available. When using the QR method, phones connected to the same network can safely send photos through a small background helper tool. It works with various types like JPG,JPEG, PNG, BMP, and TIFF, so it fits most standard eye cameras without issues.

C. *Image Preprocessing Layer*

To keep things consistent and help the model handle variations better, every retinal scan goes through an initial cleanup step. First, it finds the main eye area and cuts out extra surrounding bits that aren't needed. Then, contrast gets boosted while random specks or distortions are cleaned up. After cleaning, each picture is adjusted to fit exactly 384 × 384 pixels right for what EfficientNet-B3 needs. While learning from these images, slight flips, rotations, or brightness tweaks are added on purpose to balance under represented cases and reduce overfitting.

D. *AI Inference and Classification Layer*

The main part of the system uses an AI setup called the Inference Layer, using a smart image tool named EfficientNet-B3 to sort diabetic retinopathy levels. Instead of stacking raw pixels, it pulls out layered patterns from cleaned up eye photos spotting signs linked to one of five stages: no damage, light issues, mid level harm, intense trouble, or advanced spreading disease. With help from a scoring method based on probabilities, each result comes with a certainty level doctors can trust during review. Rather than favouring common cases, training adjusts fairly across all categories; plus, it runs fast even without graphics chips by tuning demands down for regular processors.

E. *Explainable AI Visualization Layer*

To tackle the need for clear medical AI decisions, the system uses a built in explanation tool powered by Grad-CAM. It creates heatmaps showing which parts of the retina influenced the Diabetic Retinopathy severity prediction the most. Besides that, it displays charts with likelihoods spread across all possible outcomes. These visuals help doctors check if the output makes sense, making them more likely to rely on automatic screenings.

F. *Reporting, Storage, and Communication Layer*

The last part of the setup handles how results are shown, saved, or shared. It builds PDFs in medical format that include patient info, eye scan pictures, highlight maps showing key areas, chance graphs, along with the doctor's conclusion. These files go into a local database sorted by checkup date so past visits stay organized. If needed, an email feature can send these documents safely to clinics through password protected servers. Keeping everything stored locally protects personal data while cutting reliance on outside online systems.

IV. METHODOLOGY

This part present how the Diabetic Retinopathy detection method is built. Our goal was to correctly sort Diabetic Retinopathy into multiple stages while keeping things clear and practical for everyday use. This research work started by setting up the data, then cleaned the images before feeding them into a deep learning model. Once trained, the model makes predictions using techniques that show why it made those choices. Finally, results pop out automatically in a ready to use format.

A. *Dataset Acquisition and Preparation*

The dataset was obtained from the APTOS 2019 Blindness Detection dataset and categorized into five severity classes: No DR, Mild, Moderate, Severe, and Proliferative Diabetic Retinopathy. The dataset is divided into training, validation, and testing subsets to ensure unbiased model evaluation. All images in the dataset were captured using standard fundus cameras in a clinical setting, which produce clear and well-lit retinal images. Since the proposed system is designed to work only with these fundus images, it does not

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face problems related to poor quality, blur, or occlusion, allowing the model to provide stable and reliable diagnostic results.

B. Image Preprocessing

To reduce noise and help the model work better on new each retinal fundus image undergoes a standardized preprocessing pipeline. First, the fundus region is found and cut out to skip unneeded surroundings. Methods that boost clarity are used to make signs of disease easier to see. These cleaned up images get adjusted evenly and scaled to fit 384 ×384 pixels matching what EfficientNet-B3 needs at input. While learning, extra versions such as rotation, flipping, and brightness variation are employed during training to overcome overfitting and improve robustness.

C. Deep Learning Model Architecture

The proposed system uses EfficientNet-B3 for feature extraction and classification. Instead of just making the model deeper, wider, or higher res separately, this version adjusts all three together so it's accurate but doesn't need heavy computing power. To speed things up, the network starts with weights already trained on other data, then gets adjusted specifically for diabetic retinopathy images. On top of that base, there's a tailored layer which gives likelihoods for each of the five stages of Diabetic Retinopathy severity through a SoftMax activation function.

D. Model Training and Validation

The model learns through guided examples. To handle uneven classes, it uses A weighted cross entropy loss function that employed to address class imbalance during optimization. For tuning, it picks Adam optimizer this adjusts step sizes automatically while lowering the rate over time in a smooth way, plus halts early if no improvement shows up. After every round of learning, it checks how well it's doing, then keeps the top version seen so far when tested on holdout data.

Loss Function

$$\mathcal{L} = -\sum_{i=1}^{C} w_i \, y_i \log(\hat{y}_i). \tag{1}$$

Where

C is the number of classes, y_i is the true label data,

$\hat{y}i$ is the predicted probability, And wi is the class weight.

E. Performance Evaluation Metrics

To evaluate the effectiveness of the proposed model, standard classification metrics are employed. Accuracy measures the proportion of correct predictions relative to the total number of predictions. It is defined as:

$$\text{Accuracy} = \text{TP+TN} / \text{TP+TN+FP+FN} \tag{2}$$

While accuracy provides a general overview of model performance, it does not distinguish between false positives and false negatives. Hence, Precision is used to evaluate how many of the predicted positive cases are actually relevant. It is expressed as:

$$\text{Precision} = \text{TP} / \text{TP+FP} \tag{3}$$

In contrast, Recall indicates the model's ability to correctly identify all true cases belonging to a specific Diabetic Retinopathy severity class. High recall is particularly important in diabetic retinopathy detection, where failing to identify true Diabetic Retinopathy cases may lead to delayed diagnosis. Recall is calculated as:

$$\text{Recall} = \text{TP} / \text{TP+FN} \tag{4}$$

Because both Precision and Recall capture different aspects of model performance, the F1 score defined as the harmonic mean of Precision and Recall is used to provide a balanced metric. This is especially valuable when evaluating models on datasets with class imbalance, such as those containing varying proportions of Diabetic Retinopathy severity levels:

$$\text{F1 Score} = 2 \times \text{Precision x Recall} / \text{Precision} + \text{Recall} \tag{5}$$

These metrics provide a comprehensive evaluation of model performance across all Diabetic retinopathy severity classes.

F. Explainable AI Using Grad-CAM

To make things clearer and more trustworthy in practice, the new system uses something called Grad-CAM. Instead of just guessing, it creates heatmaps using gradient data from the last convolutional layer showing which parts of the retina affect the Diabetic Retinopathy prediction most. This way, doctors can spot problem areas like bleeding spots or tiny bulges easier because they see what the model focused on. It helps them decide better since they're not working blind.

G. Algorithmic Workflow

Algorithm: Proposed Diabetic Retinopathy Detection Method

1. Get a retinal image by sending a file, using a mobile scan, or webcam.
2. Clean up the picture crop the eye part, normalize, resize.
3. Push the cleaned image into EfficientNet-B3 for processing.
4. Predict Diabetic Retinopathy severity level along with how sure the system is about it.
5. Create a Grad-CAM heatmap using model output then map out the likelihood spread across classes.
6. Show test results, then create a printable health summary.
7. Share the report by email if patient like just make sure it's secure.

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H. Implementation Details

The setup runs on Python, using PyTorch to handle deep learning tasks. For the front end, Streamlit powers the web interface, whereas Flask manages mobile uploads through QR codes. Visuals like Grad-CAM come from standard plotting tools, meanwhile ReportLab handles building PDF summaries. It works on regular CPUs, also offers Docker packaging so it's easier to move between machines.

The suggested method combines fair data practice with EfficientNet-B3 sorting, clear AI visuals.

V. Result and Discussion

This section presents the performance of the Diabetic retinopathy system built using EfficientNet-B3, while exploring how well the model works through various scoring methods. The model was evaluated using different metrics such as accuracy, precision, recall, F1-score, ROC AUC, and Quadratic Weighted Kappa to understand how well it performs across all disease stages.

A. Dataset Distribution and Balancing

Initially, the dataset contained many more normal and mild cases, while severe and advanced stages were much fewer. This imbalance could cause the model to favor common cases and miss serious conditions. To overcome this, the dataset was balanced during training so that all five severity levels were represented more equally. As a result, rare classes like Severe and Proliferative Diabetic Retinopathy increased significantly, allowing the model to learn more fairly from all categories.

B. Classification Performance and Per Class Evaluation

The EfficientNet-B3 approach hit a 98.77% accuracy rate on the test data, showing strong results for every stage of diabetic retinopathy. Performance scores by category also back up how well this setup works. The accuracy, recall, f1 results stay solid for every one of the five groups. Not sick and mid level eye damage spots hit almost flawless scores showing clear separation between normal and damaged retina scans. What's key, serious and worst stage issues had strong catch rates too, which matters a lot since skipping these could cause permanent sight problems. A strong average F1-score indicates robust performance, even when certain categories contain fewer samples.

TABLE I. Classification Performance Metrics for Diabetic Retinopathy Severity Levels

Overall Accuracy: 0.9877 (98.77%)				
	precision	recall	f1-score	support
No DR	1.0000	0.9922	0.9961	Value 3
Mild	0.9385	1.0000	0.9683	386
Moderate	1.0000	0.9730	0.9863	61
Severe	0.9211	1.0000	0.9589	35
Proliferative DR	0.9697	0.9846	0.9771	65
accuracy			0.9877	732
macro avg	0.9658	0.9900	0.9773	732
weighted avg	0.9884	0.9877	0.9878	732

C. ROC AUC Analysis for Multi Class Classification

This Figure shows the ROC curves for all five Diabetic Retinopathy categories. The curves are positioned very close to the top-left corner, showing that the model can clearly separate each disease stage from the others. The AUC values for No DR, Mild, Moderate, Severe, and Proliferative classes are all nearly 1.0, which indicates excellent discrimination performance. The dashed diagonal line represents a random classifier, while all model curves remain far above this baseline, proving that the predictions are not based on chance. These strong ROC–AUC results highlight the ability of the EfficientNet-B3 model to capture important retinal patterns and accurately distinguish between different levels of Diabetic Retinopathy severity Fig 5.

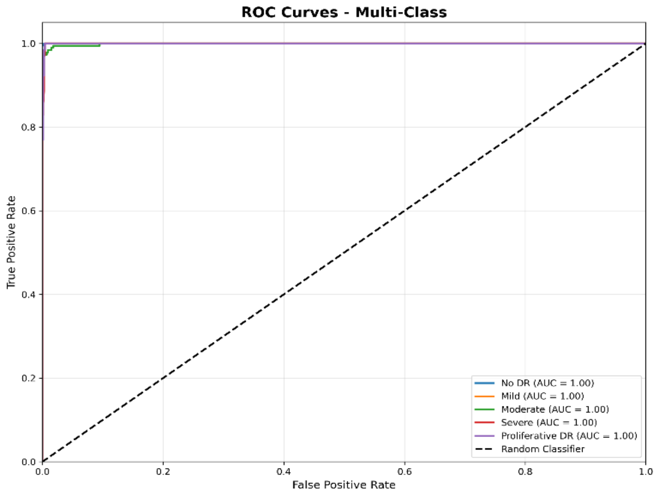

Fig. 5. Multi Class ROC Curves for Diabetic Retinopathy Severity Classification

D. F1 Score Convergence and Quadratic Weighted Kappa

The F1 score changes during training reveal how well the model settles and learns over time. While the macro version rises steadily across epochs, so does the weighted one suggesting better performance and adaptation. Each step forward helps fine tune results without major hiccups along the way. The way things improved shows the training tricks like adjusting class sizes, using a smooth learning drop off. The QWK score not just accuracy to check how well predictions matched real labels, especially since diabetes related eye damage gets worse step by step. Instead of treating each stage separately, this method considers that some mistakes matter more than others like calling mild disease severe versus missing it entirely. The validation kappa went up step by step through epochs hitting a max of 0.9914 sign of near total alignment.

E. Effect of Grad-CAM on Model Performance

Grad-CAM in this system is used only to visually show which parts of the eye image influenced the model's prediction. The EfficientNet-B3 model first predicts the stage of diabetic retinopathy, and then Grad-CAM creates heatmaps to show which parts of the retinal image were important for that prediction. Since Grad-CAM is applied

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after the prediction stage, it does not participate in or alter the classification process. Therefore, disabling Grad-CAM does not affect the model's accuracy, F1-score, or prediction results. Whether Grad-CAM is enabled or disabled, the core EfficientNet-B3 model maintains the same diagnostic performance, while Grad-CAM only adds an optional layer of visual explanation for clinicians.

F. Computational Complexity and CPU-Based Inference Performance

The proposed system was designed to run efficiently in CPU-only environments, making it suitable for real world medical use. The EfficientNet-B3 architecture provides a good balance between model size and accuracy, allowing fast and stable inference without the need for expensive GPU hardware. During testing, the system performed reliably on standard CPUs and produced predictions without delay, even when integrated with preprocessing, visualization, and report generation modules. Despite operating on CPU, the model achieved a high overall accuracy of 98.77%, demonstrating that strong diagnostic performance can be maintained with low computational cost.

G. Class Balancing and Minority Class Performance

The APTOS 2019 dataset used in this study contains an uneven distribution of images across the five diabetic retinopathy stages. To handle this imbalance, weighted sampling and a weighted loss function were applied during training so that under-represented classes, such as Severe and Proliferative DR, received greater importance during learning. This prevented the model from being biased toward more common classes like No DR and Mild DR. As a result, the system was able to recognize advanced disease stages more effectively, achieving a high F1-score of 0.958 for Severe DR, along with strong performance for other minority classes. These results confirm that the class balancing strategy significantly improved prediction reliability across all severity levels, making the system suitable for real-world screening.

H. Comparison with State of the Art Methods

The proposed EfficientNet-B3 model was compared with commonly used deep learning architectures such as CNN, ResNet-50, DenseNet-121, and Inception-V3 using the same APTOS-2019 diabetic retinopathy dataset. Among these models, EfficientNet-B3 achieved the highest performance with an overall accuracy of 98.77%, showing better classification ability across all disease stages. This comparison demonstrates that the proposed model outperforms existing state-of-the-art methods for diabetic retinopathy detection when evaluated on the same dataset.

TABLE II. PERFORMANCE COMPARISON WITH STATE OF THE ART MODELS

Model	Dataset	Accuracy	F1-Score
CNN	APTOS-2019	91.4%	0.90
ResNet_50	APTOS-2019	94.6%	0.93
DenseNet-121	APTOS-2019	95.2%	0.94
Inception-V3	APTOS-2019	96.1%	0.95
EfficientNet-B3	APTOS-2019	98.77%	0.98

I. Discussion and Clinical Implications

The tests show the Diabetic Retinopathy works really well, yet still runs fast enough for everyday use. Because of balanced data, an EfficientNet-B3 setup, also tools that clarify decisions. On top of that, having access to Grad-CAM heatmaps along with probability charts makes it easier to understand what the AI sees helping ophthalmologists check results at a glance. Combined with auto generated reports and support for various image uploads, this setup connects lab findings with real world medical practice.

VI. CONCLUSION

This study presented a complete system for detecting and grading Diabetic Retinopathy from retinal images using the EfficientNet-B3 model. The system classifies images into five disease stages, from healthy to advanced conditions, using well-prepared and balanced data to ensure fair learning across all categories. Users can upload images through file selection, QR-based mobile access, or live camera capture, making the platform practical for real-world use. The model achieved a high accuracy of 98.77%, along with strong precision, recall, and F1-scores for all classes. The confusion matrix, ROC–AUC curves, and Quadratic Weighted Kappa results confirm that the predictions closely match actual disease labels. Data balancing also helped reduce bias toward common cases, improving reliability for severe stages. Grad CAM visual maps highlight important regions in the retinal images, helping doctors understand how decisions are made and increasing trust in the system. With features like automatic PDF report generation and email sharing, the platform fits well into tele-ophthalmology and clinic workflows. Overall, this solution provides an accurate, transparent, and easy-to-use tool for early detection of Diabetic Retinopathy, helping reduce specialist workload and support timely treatment to prevent vision loss.

REFERENCES

[1] L. Sivayamini, C. Venkatesh, B. Chalivendula, G. H. P. Annadanam, A. Kanaganipalli, and C. S. Mogili, "A dynamic deep learning model for detection and stage categorization of diabetic retinopathy on retinal images," in *Proc. 2024 Int. Conf. Recent Innovation in Smart and Sustainable Technology (ICRISST)*, Mar. 2024, pp. 1–5.

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[2] S. Khanapur and L. Patil, "A study on diabetic retinopathy using deep learning algorithms," in *Proc. 2023 Int. Conf. Integrated Intelligence and Communication Systems (ICIICS)*, Nov. 2023, pp. 1–5.

[3] S. Muntaha, S. S. Salam, and N. Mustakim, "An explainable AI-based deep learning model for classification of diabetic retinopathy stages using retinal fundus images," in *Proc. 27th Int. Conf. Computer and Information Technology (ICCIT)*, Dec. 2024, pp. 3010– 3015.

[4] A. Matthew, A. A. S. Gunawan, and F. I. Kurniadi, "Diabetic retinopathy diagnosis system based on retinal biomarkers using EfficientNet-B0 for Android devices," in *Proc. 2023 IEEE Int. Conf. Communication, Networks and Satellite (COMNETSAT)*, Nov. 2023, pp. 207–212.

[5] M. Asif, F. ur Rehman, Z. Rashid, A. Hussain, A. Mirza, and W. S. Qureshi, "An insight on the timely diagnosis of diabetic retinopathy using traditional and AI-driven approaches," *IEEE Access*, 2025.

[6] K. Upreti, A. Kapoor, S. Hundekari, S. Upreti, K. Kaul, S. Kapoor, and A. Tiwari, "Deep dive into diabetic retinopathy identification: A deep learning approach with blood vessel segmentation and lesion detection," *J. Mobile Multimedia*, vol. 20, no. 2, pp. 495–523, 2024.

[7] A. Zedadra, O. Zedadra, M. Y. Salah-Salah, and A. Guerrieri, "Graph-aware multimodal deep learning for classification of diabetic retinopathy images," *IEEE Access*, 2025.

[8] S. Zhu, C. Xiong, Q. Zhong, and Y. Yao, "Diabetic retinopathy classification with deep learning via fundus images: A short survey," *IEEE Access*, vol. 12, pp. 20540–20558, 2024.

[9] K. A. Alavee, M. Hasan, A. H. Zillanee, M. Mostakim, J. Uddin, E. S. Alvarado, and M. A. Samad, "Enhancing early detection of diabetic retinopathy through the integration of deep learning models and explainable artificial intelligence," *IEEE Access*, vol. 12, pp. 73950–73969, 2024.

[10] A. M. Mutawa, G. R. Hemalakshmi, N. B. Prakash, and M. Murugappan, "Randomization-driven hybrid deep learning for diabetic retinopathy detection," *IEEE Access*, vol. 13, pp. 38901–38913, 2025.

[11] S. K. Henge, N. R. Viraati, M. Alhussein, A. S. Kushwaha, K. Aurangzeb, and R. Singh, "Detection of diabetic retinopathy using a multi-decision Inception– ResNet-blended hybrid model," *IEEE Access*, 2025.

[12] I. Y. Abushawish, S. Modak, E. Abdel-Raheem, S. A. Mahmoud, and A. J. Hussain, "Deep learning in automatic diabetic retinopathy detection and grading systems: A comprehensive survey and comparison of methods," *IEEE Access*, vol. 12, pp. 84785–84802, 2024.

[13] S. J. Sidiq and T. Benil, "A lightweight transfer learning based ensemble approach for diabetic retinopathy detection," *Int. J. Information Management Data Insights*, vol. 5, no. 2, p. 100372, 2025.

[14] http://naveeneyehospital.com/services/diabetic-retinopathy/

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